

Closure-Corrected Cosmology: Plateau and Inflection Signatures Beyond Λ CDM

Abstract

We investigate a closure-corrected extension of Λ CDM cosmology in which deviations arise from a structural completion parameter $C(a)$, with closure deficit $\varepsilon(a) = 1 - C(a)$.

The effective dark energy term acquires a controlled dynamic component proportional to $\varepsilon(a)$. A quasi-plateau regime naturally emerges when structural parameters satisfy $\beta \approx \gamma$. The model preserves the Λ CDM backbone and recovers it in the completion limit $C \rightarrow 1$.

This work proposes a conservative phenomenological extension that does not modify General Relativity but introduces a parameter-controlled correction term with explicit recovery behavior.

1. Introduction

Λ CDM remains the dominant cosmological model, yet the nature of dark energy remains unknown. Small deviations in $w(z)$ remain observationally allowed.

Rather than modifying gravity or introducing new scalar fields, we consider a minimal structural correction framework in which deviations arise from a closure deficit $\varepsilon = 1 - C$.

The goal is not to replace Λ CDM but to parameterize controlled deviations while preserving the standard Friedmann structure.

2. Closure Dynamics

We define structural evolution as:

$$dC/d(\ln a) = (1 - C)(\beta - \gamma C)$$

Properties:

- Fixed points: $C = 1$ and $C = \beta/\gamma$
- Inflection: $C = \beta/(2\gamma)$

Closure deficit:

$$\varepsilon(a) = 1 - C(a)$$

3. Modified Friedmann Equation

Effective dark energy term:

$$\Lambda_{\text{eff}}(a) = \Lambda_0 + \Lambda_1 \varepsilon(a)$$

Normalized density:

$$\Omega_{\Lambda\text{cl}}(a) = \Omega_{\Lambda\text{cl},0} * \varepsilon(a) / \varepsilon(1)$$

Modified Friedmann equation:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_{\Lambda 0} + \Omega_{\Lambda\text{cl}}(a)]$$

Limit:

$$C \rightarrow 1 \Rightarrow \Omega_{\Lambda\text{cl}} \rightarrow 0$$

Standard Λ CDM recovered exactly.

4. Effective Equation of State

$$w_{\text{eff}}(a) = -1 - (1/3) d \ln \rho_{\text{DE}} / d \ln a$$

Near completion:

$$d \ln \varepsilon / d \ln a \approx -(\beta - \gamma)$$

Thus:

$$w_{\text{eff}} \approx -1 + (1/3)(\Omega_{\Lambda\text{cl}} / (\Omega_{\Lambda 0} + \Omega_{\Lambda\text{cl}}))(\beta - \gamma)$$

$\beta \approx \gamma \Rightarrow$ quasi-plateau regime.

5. Parameter-Space Classification

Region I: $\beta/\gamma < 0.9$

Rapid decay of correction; nearly indistinguishable from Λ CDM.

Region II: $0.9 \leq \beta/\gamma \leq 1.1$

Quasi-plateau regime with small persistent deviation.

Region III: $\beta/\gamma > 1.1$

Enhanced late-time deviation with inflection structure.

6. Stability and Consistency

The model:

- Does not modify Einstein equations
- Introduces no new propagating degrees of freedom
- Preserves matter coupling

- Recovers Λ CDM exactly in the completion limit

This extension is phenomenological and does not claim to resolve the Hubble tension or provide a fundamental theory of dark energy.

7. Conclusion

We present a minimal closure-corrected cosmological extension with:

- One structural parameter
- Exact Λ CDM recovery
- Controlled deviation signatures
- Explicit parameter classification

Future work includes parameter fitting to cosmological datasets and numerical comparison with observational constraints.